Risk factors of severe COVID-19 in multiple sclerosis patients: A Systematic Review and Meta-Analysis: Supplementary Material

1. Forest Plots

The following figures are forest plots, showing the extracted results from each individual study, I-squared and Cochran's Q and the method used for meta-analysis.

Abbreviations:

exp(b): exponentiate of beta (Odds Ratio)

CI: confidence interval

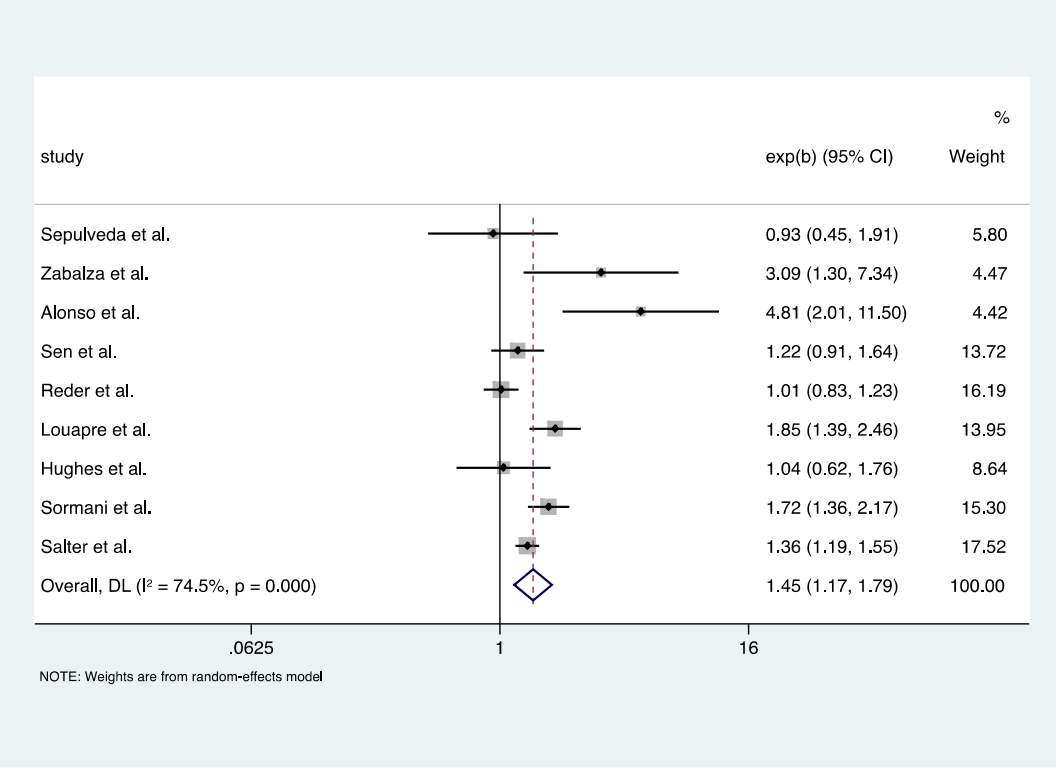


Figure 1; Age (OR per 10 years)

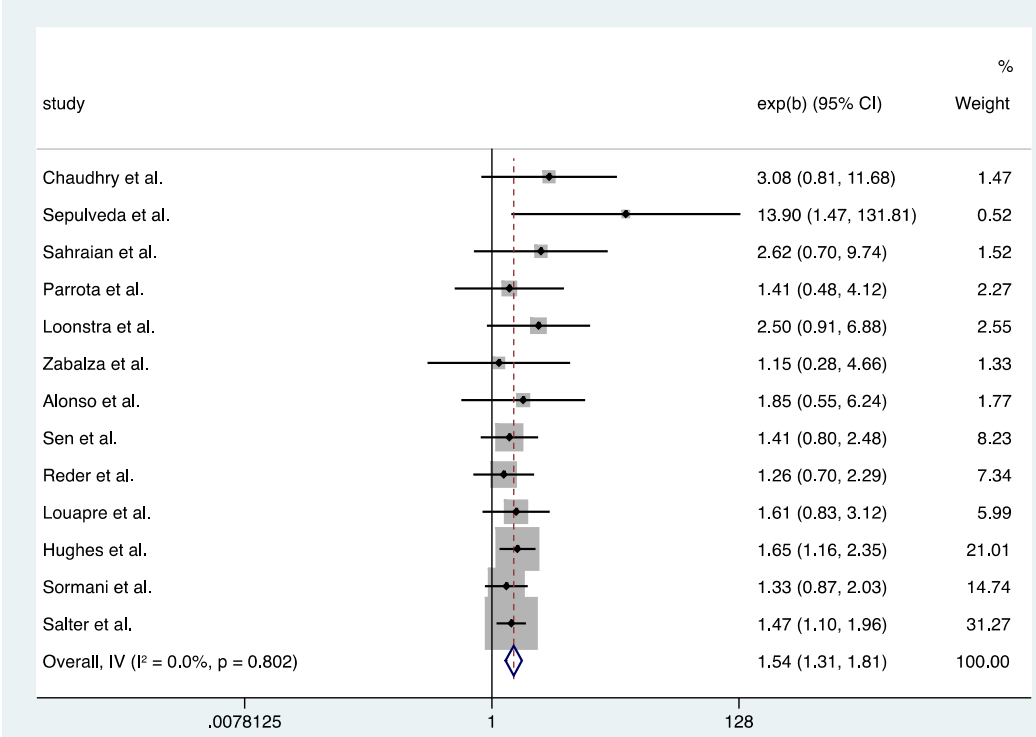


Figure 2; Male sex (OR vs. female)

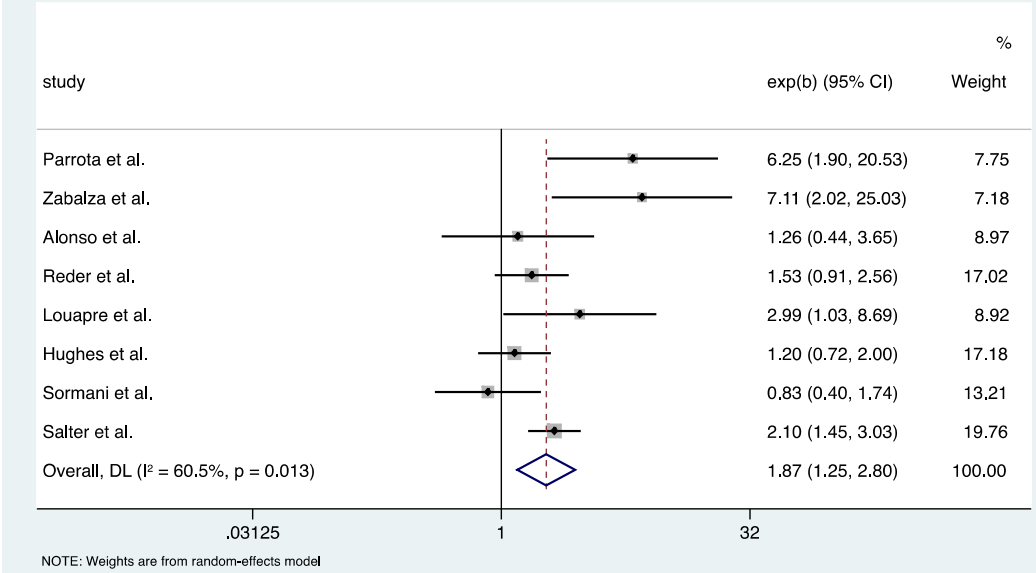


Figure 3; Obesity (OR vs. non-obese)

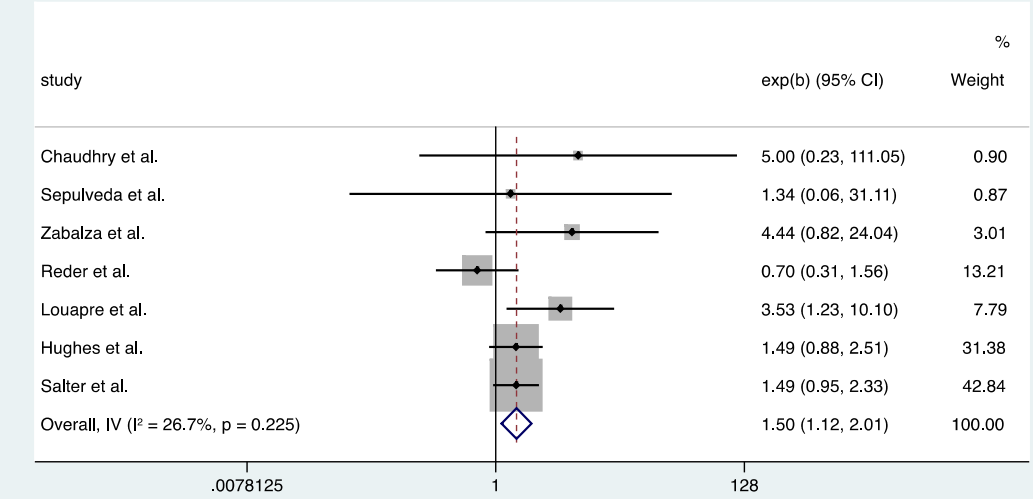


Figure 4; Pulmonary comorbidities (OR vs. no pulmonary comorbidities)

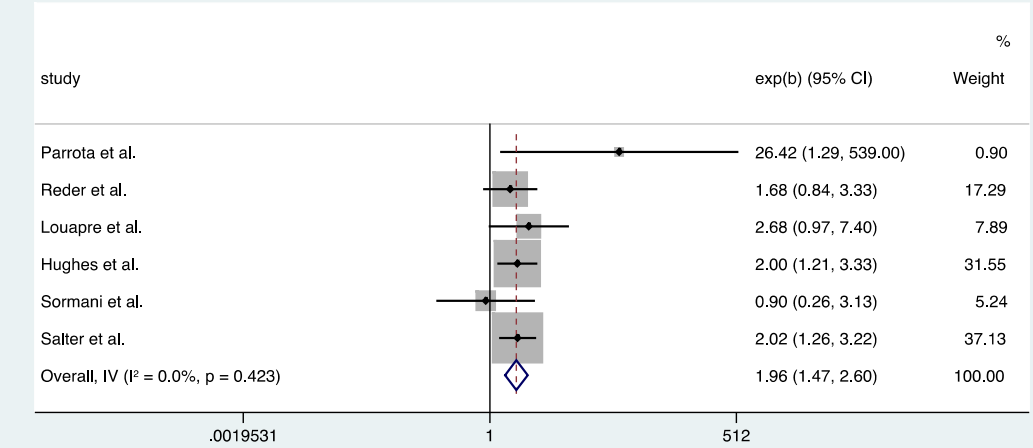


Figure 5; Cardiovascular comorbidities (OR vs. no cardiovascular comorbidities)

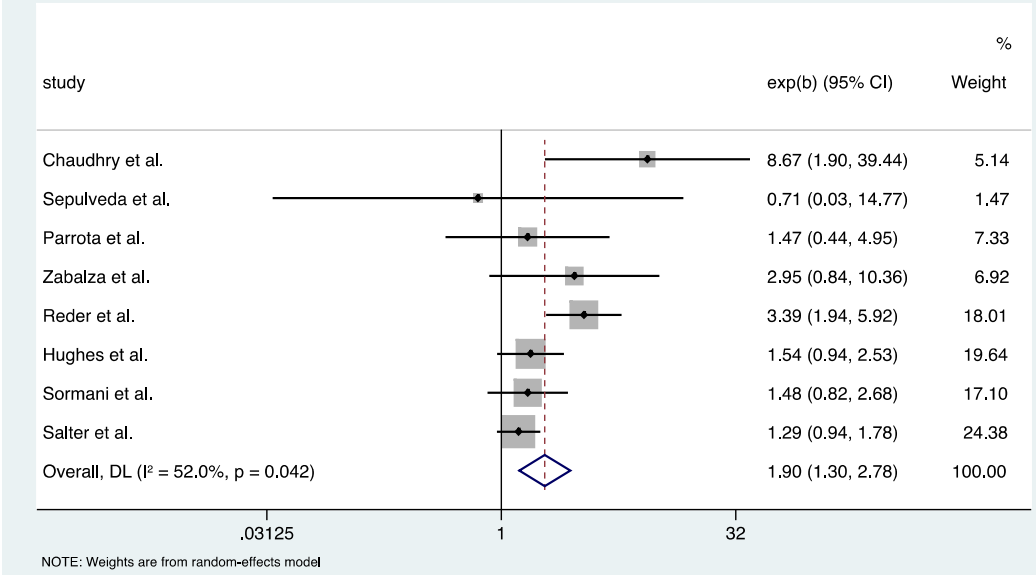


Figure 6; Hypertension (OR vs. non-hypertensive)

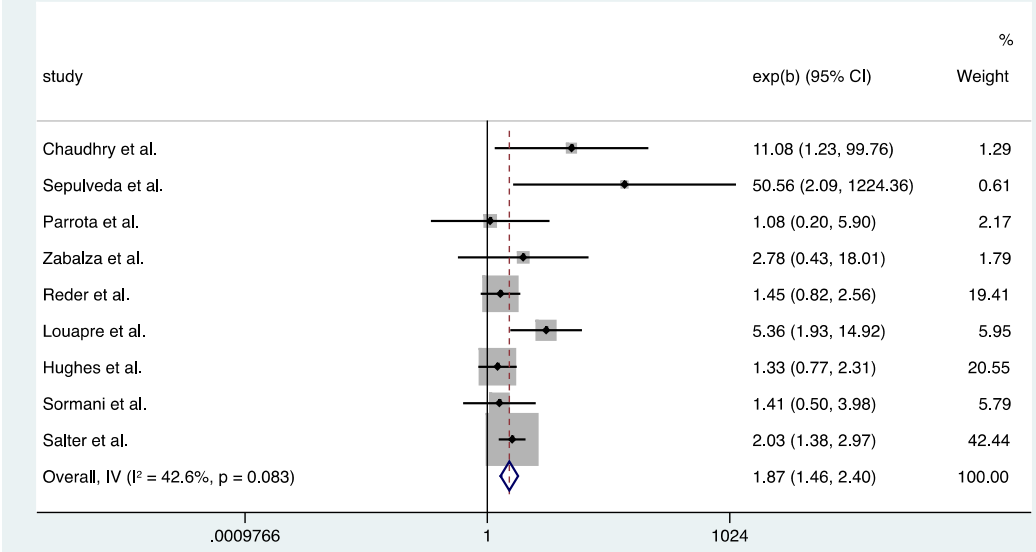


Figure 7; Diabetes (OR vs. non-diabetic)

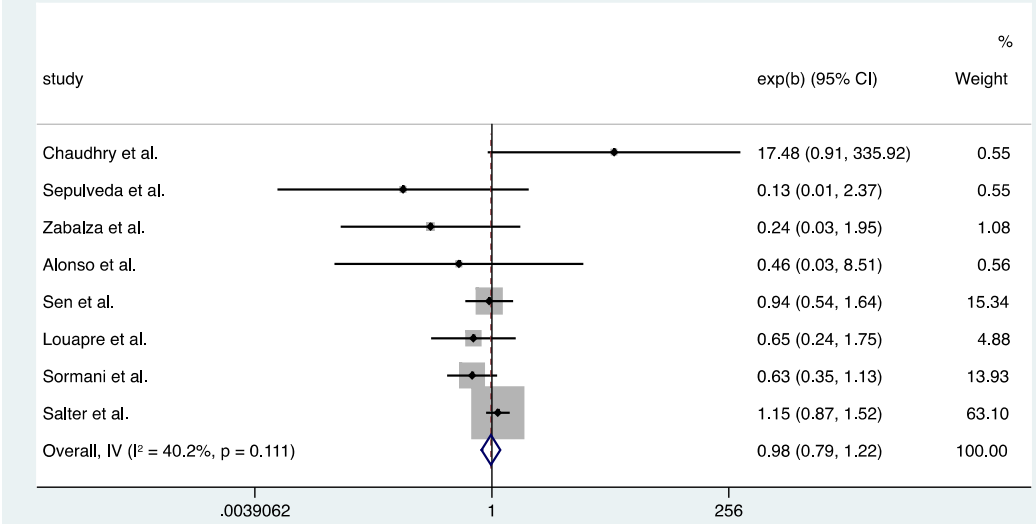


Figure 8; Smoking (OR vs. never-smokers)

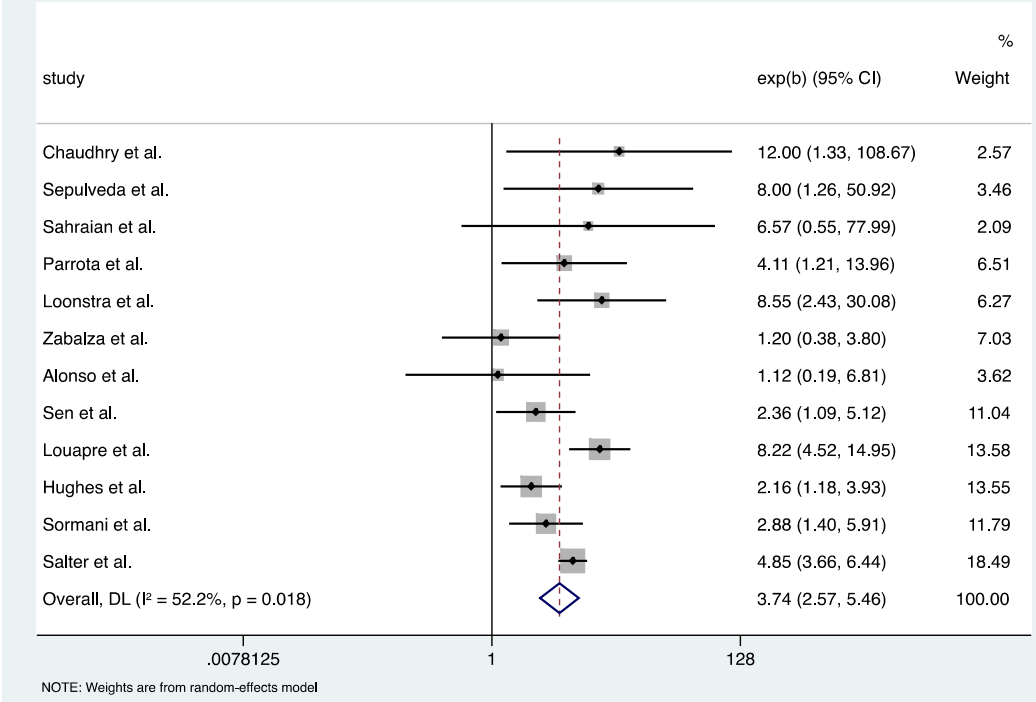


Figure 9; Progressive MS (OR vs. non-progressive)

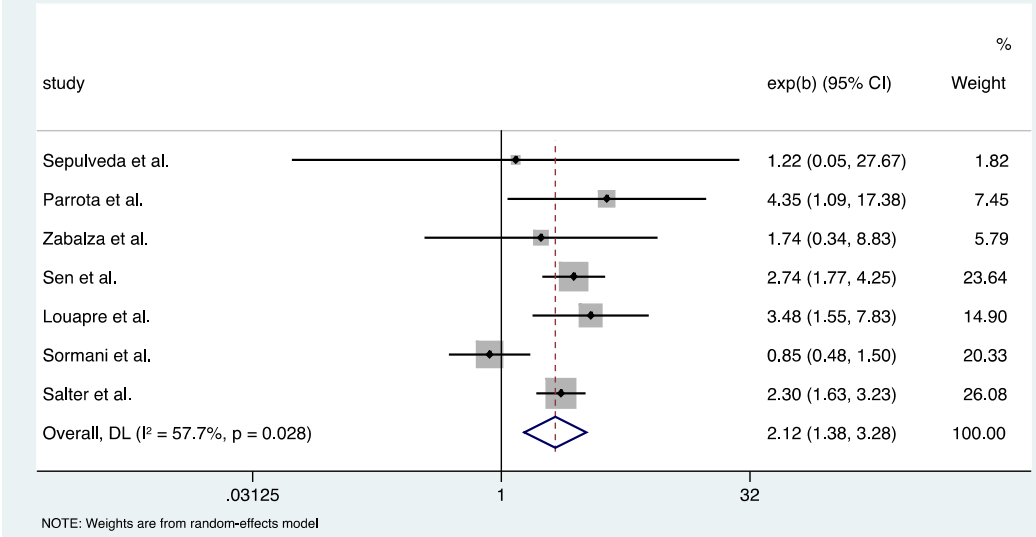


Figure 10; 3<EDSS<6 (OR vs. EDSS<3)

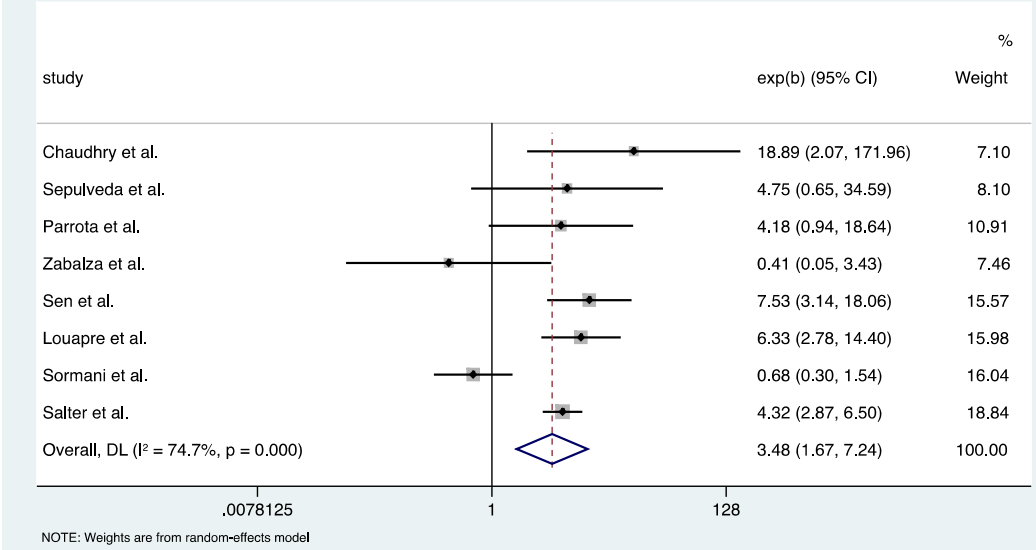


Figure 11; EDSS>6 (OR vs. EDSS<3)

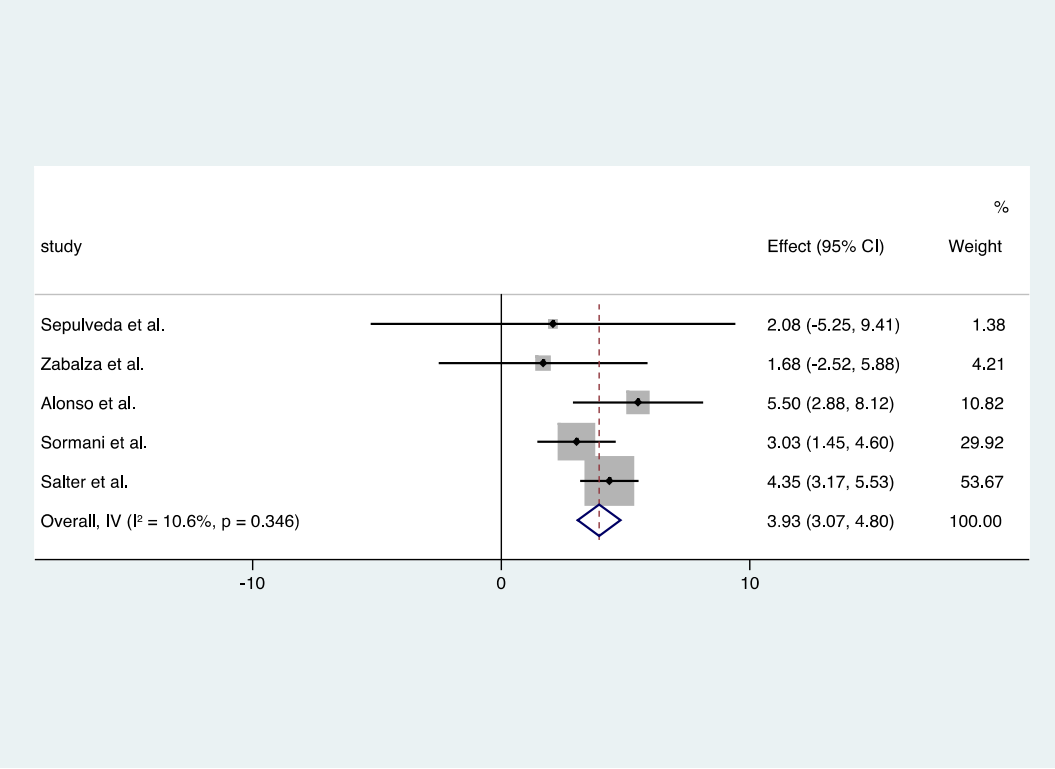


Figure 12; Disease duration (mean difference of hospitalized vs. non-hospitalized)

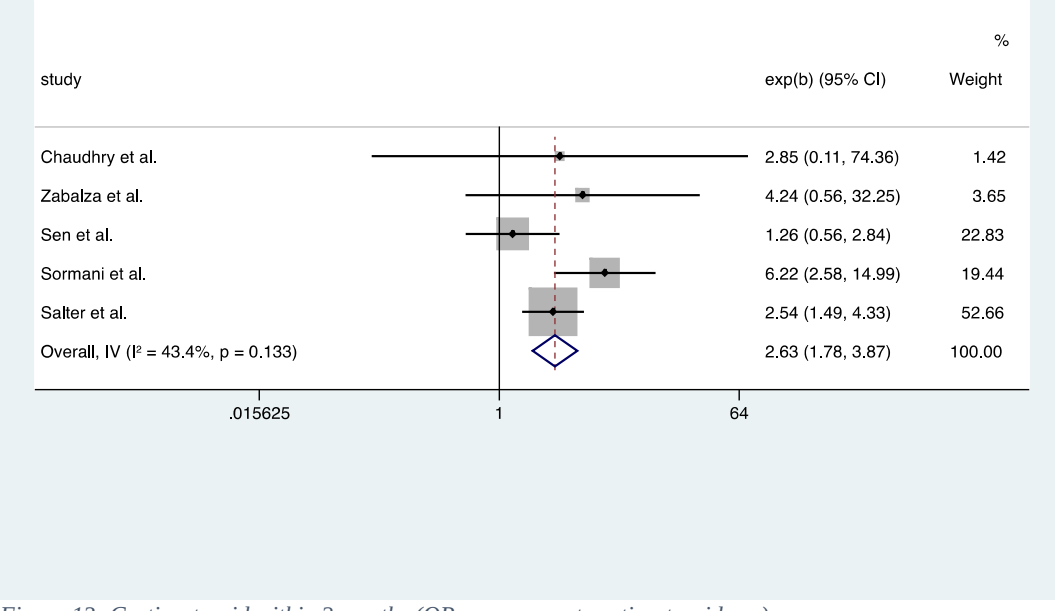


Figure 13; Corticosteroid within 2 months (OR vs. no recent corticosteroid use)

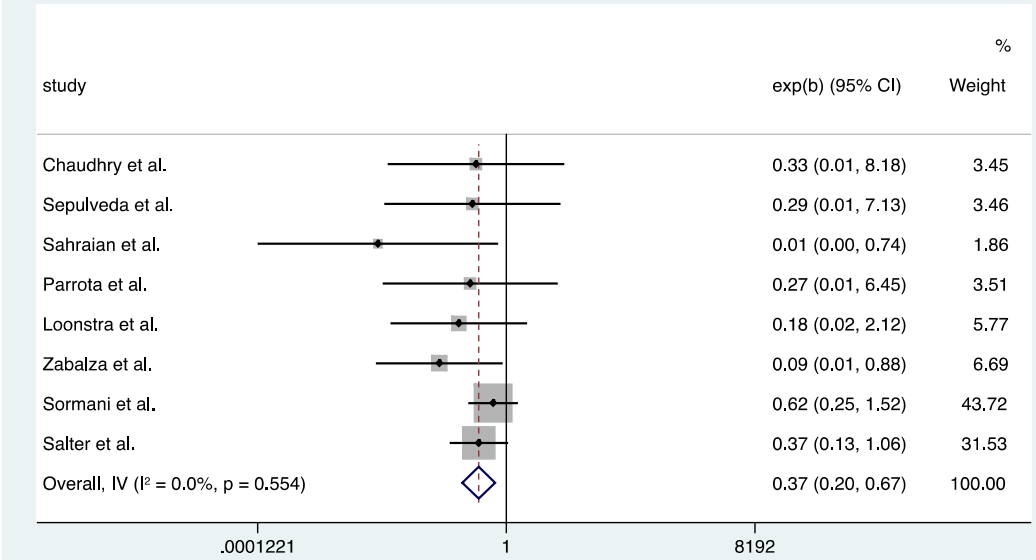


Figure 14; Interferons (OR vs. no DMTs)

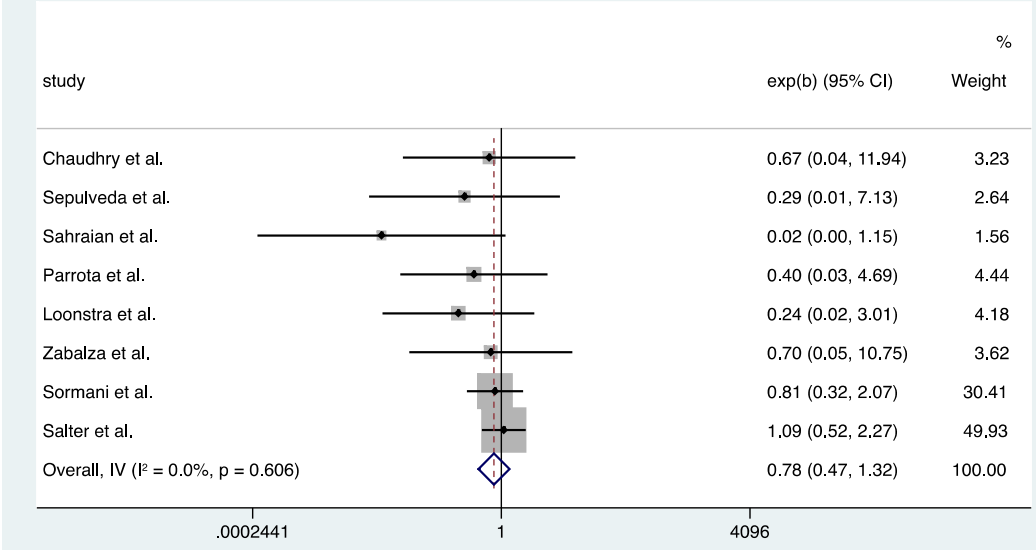


Figure 15, Glatiramer (OR vs. no DMTs)

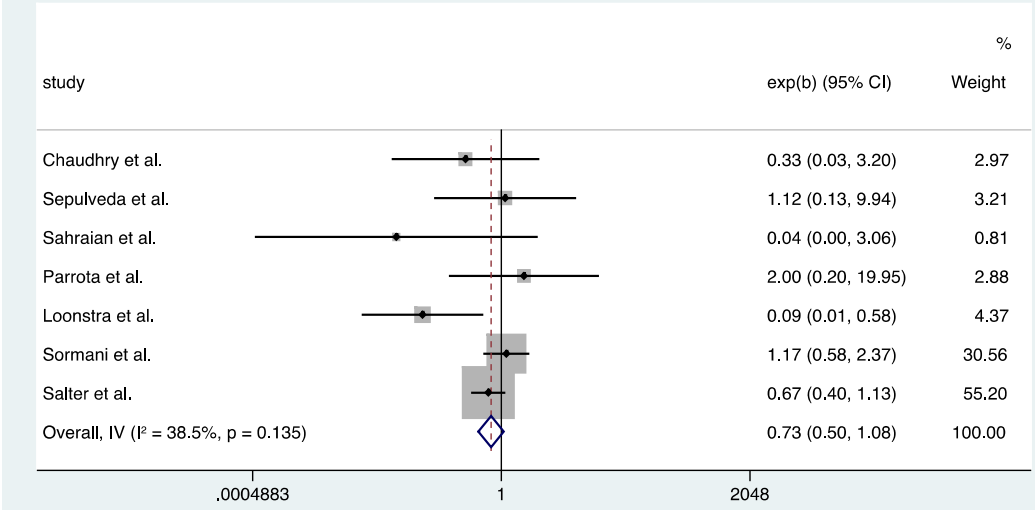


Figure 16; Fumarate (OR vs. no DMTs)

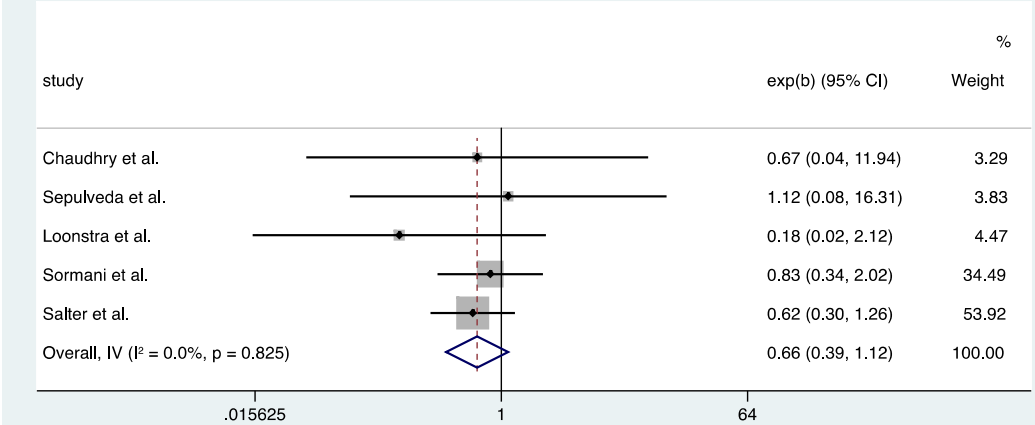


Figure 17; Teriflunomide (OR vs. no DMTs)

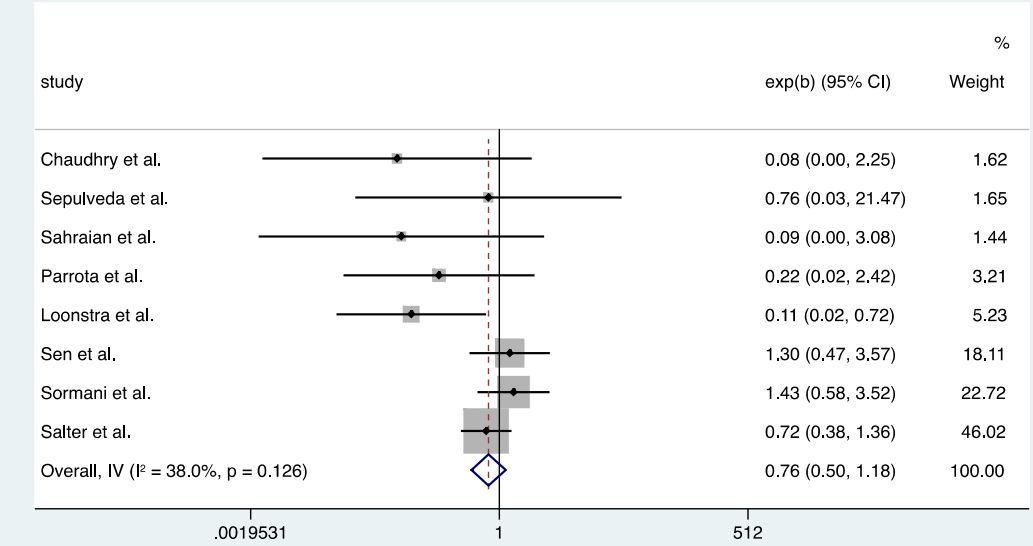


Figure 18; Fingolimod (OR vs. no DMTs)

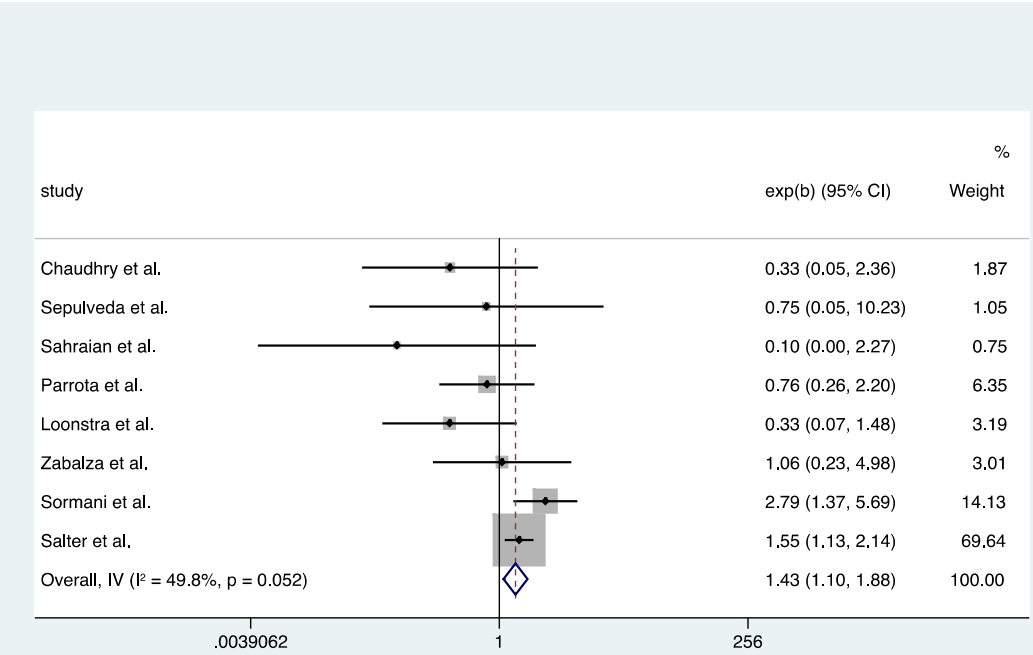


Figure 19; Anti-CD20 therapies (OR vs. no DMTs)

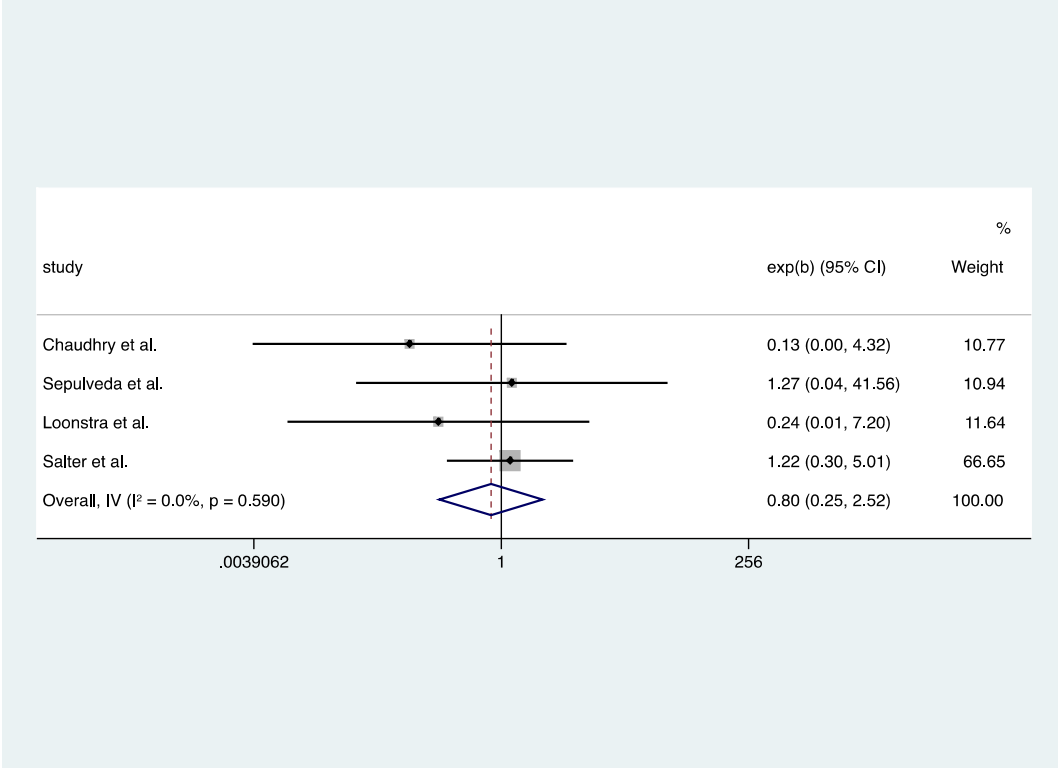


Figure 20; Alemtuzumab (OR vs. no DMTs)

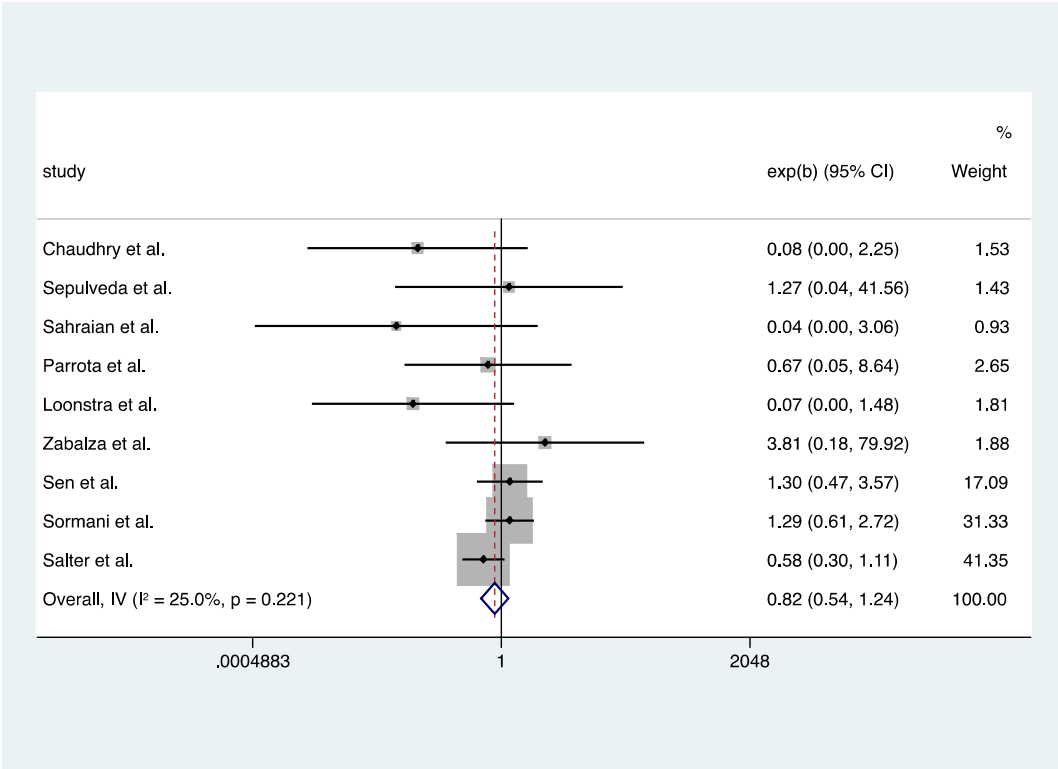


Figure 21; Natalizumab (OR vs. no DMTs)

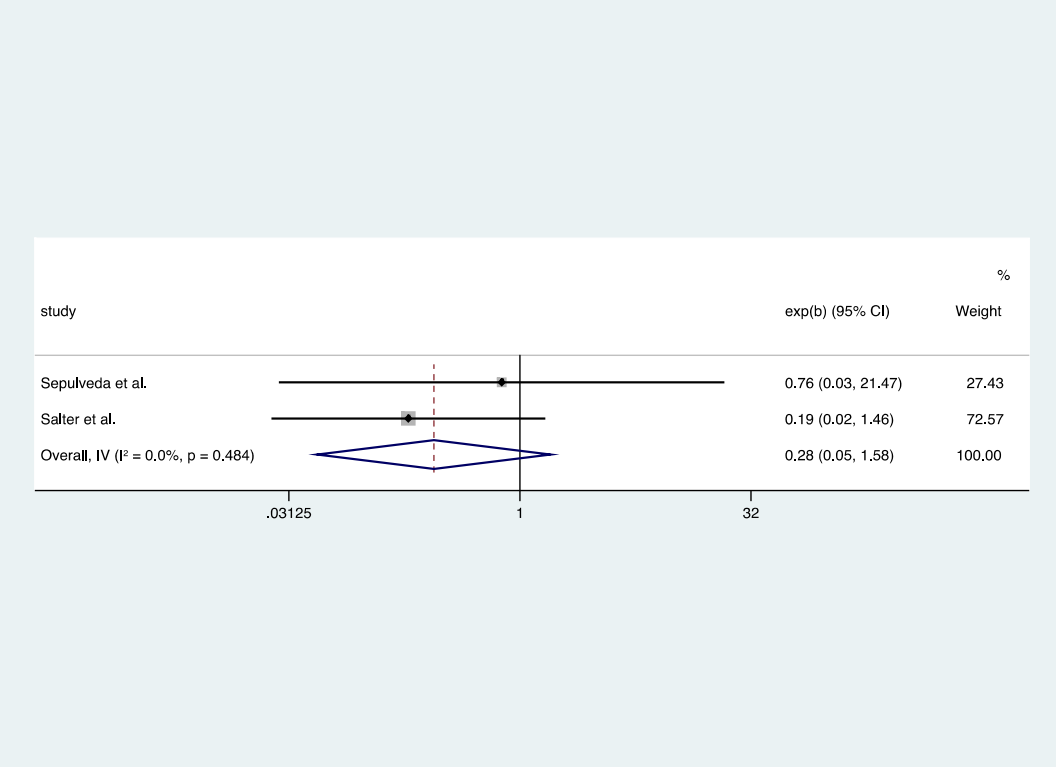


Figure 22; Cladribine (OR vs. no DMTs)

2. Quality Assessment Results, Based on NIH Quality Assessment for Observational Cohort and Cross-Sectional Studies

Study	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Quality
Alonso et al.	yes	yes	N/A	yes	No	no	N/A	yes	yes	N/A	yes	N/A	N/A	No	Fair
Brum et al.	yes	Yes	Yes	Yes	No	Yes	no	CD	yes	no	Yes	no	CD	No	Fair
Chaudhry et al.	Yes	Yes	N/A	Yes	No	No	N/A	Yes	Yes	N/A	Yes	N/A	No	No	Fair
Hughes et al.	Yes	Yes	CD	No	No	No	N/A	Yes	Yes	N/A	Yes	N/A	N/A	No	Fair
Kovvuru et al.	Yes	yes	Yes	yes	No	no	n/a	yes	Yes	No	Yes	n/a	No	No	Fair
Loonstra et al.	Yes	Yes	n/a	CD	No	No	n/a	Yes	Yes	no	Yes	n/a	n/a	No	Fair
Louapre et al.	Yes	Yes	Yes	Yes	Yes	no	n/a	Yes	Yes	n/a	Yes	n/a	n/a	Yes	Good

Parotta et al.	Yes	Yes	n/a	Yes	No	No	n/a	No	Yes	No	Yes	n/a	No	No	Fair
Reder et al.	Yes	Yes	n/a	Yes	No	No	n/a	Yes	Yes	n/a	Yes	n/a	n/a	No	Fair
Sahraian et al.	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes	no	Yes	n/a	no	No	Fair
Salter et al.	Yes	Yes	n/a	Yes	Yes	No	n/a	n/a	yes	n/a	yes	n/a	n/a	Yes	Good
Sen et al.	Yes	Yes	n/a	Yes	No	No	n/a	Yes	Yes	No	Yes	n/a	n/a	Yes	Fair
Sepulveda et al.	Yes	Yes	Yes	Yes	No	No	n/a	Yes	Yes	n/a	Yes	n/a	n/a	No	Fair
Sormani et al.	Yes	Yes	n/a	Yes	Yes	no	n/a	n/a	Yes	No	Yes	n/a	n/a	Yes	Good
Zabalza et al.	Yes	Yes	CD	Yes	No	No	n/a	Yes	Yes	No	Yes	n/a	n/a	No	Fair

3. Funnel Plots

The following figures are funnel plots, used to assess the risk of missing results (publication/reporting bias). The orange line corresponds to egger's regression line. Ideally, this line would fit on the central line of the funnel plot. This line usually points to the direction of the real effect size.

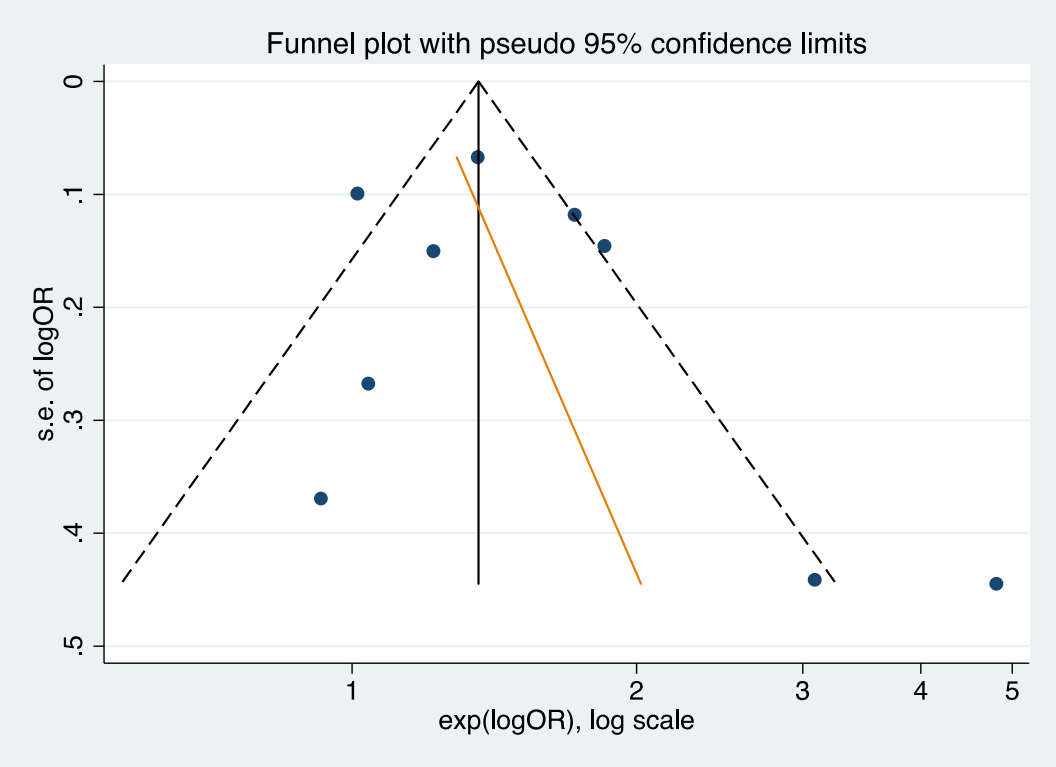


Figure 23; Age

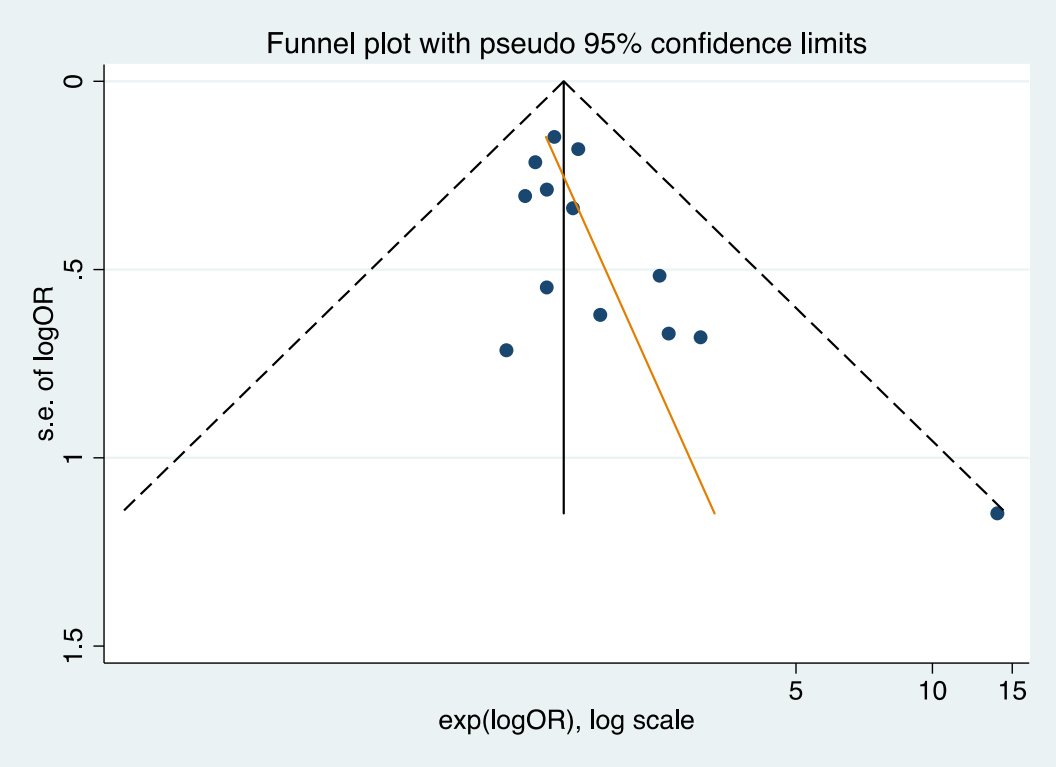


Figure 24; Male sex

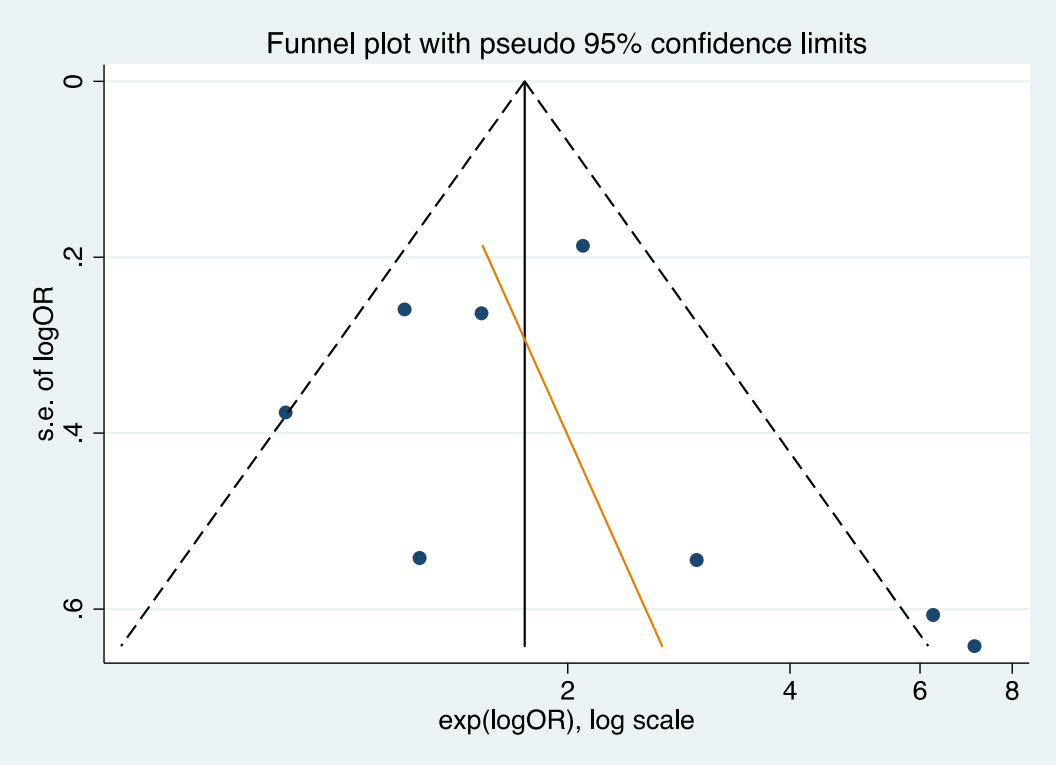


Figure 25; Obesity

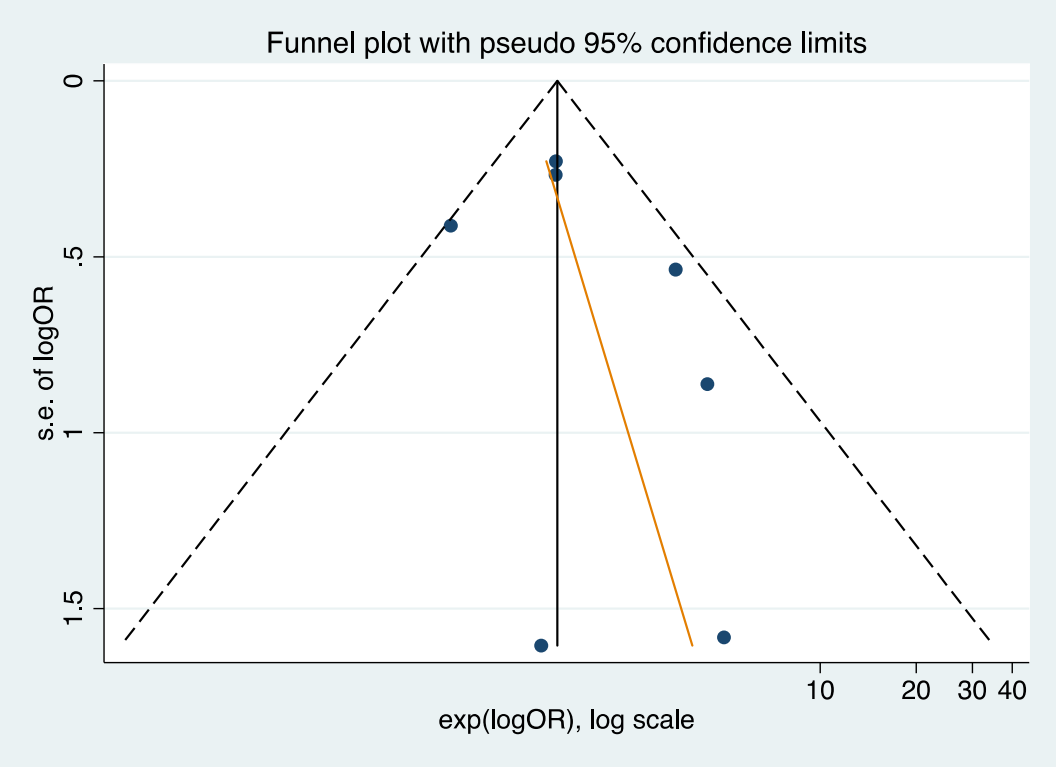


Figure 26; Pulmonary comorbidity

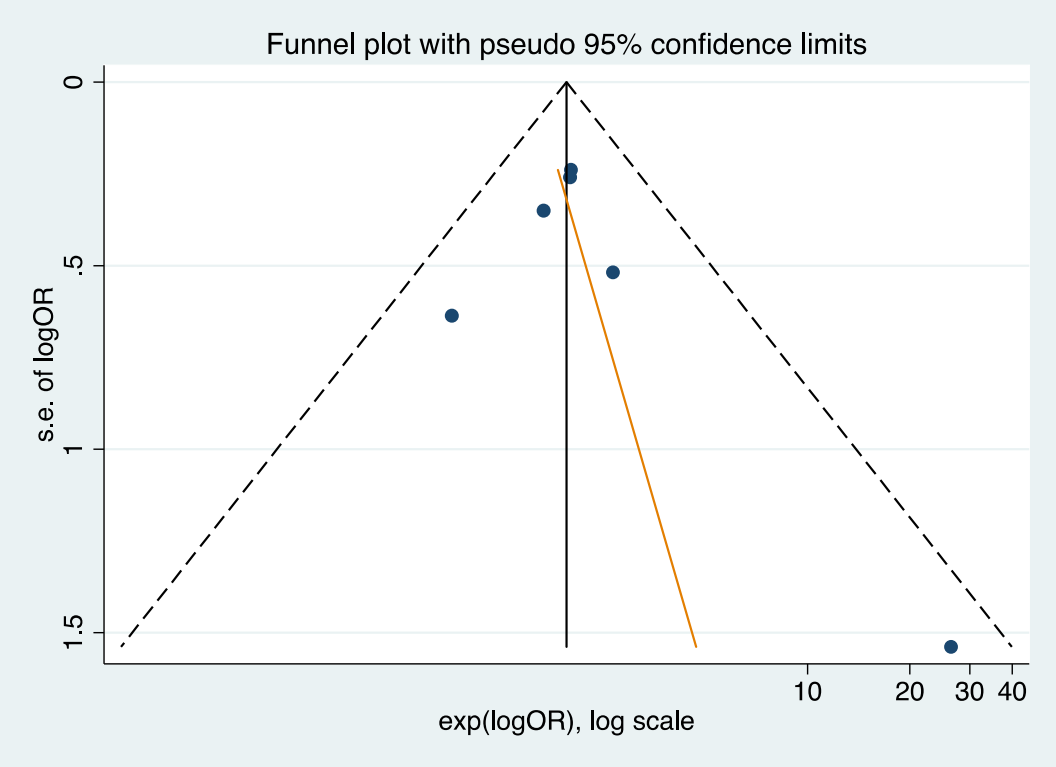


Figure 27; Cardiovascular comorbidity

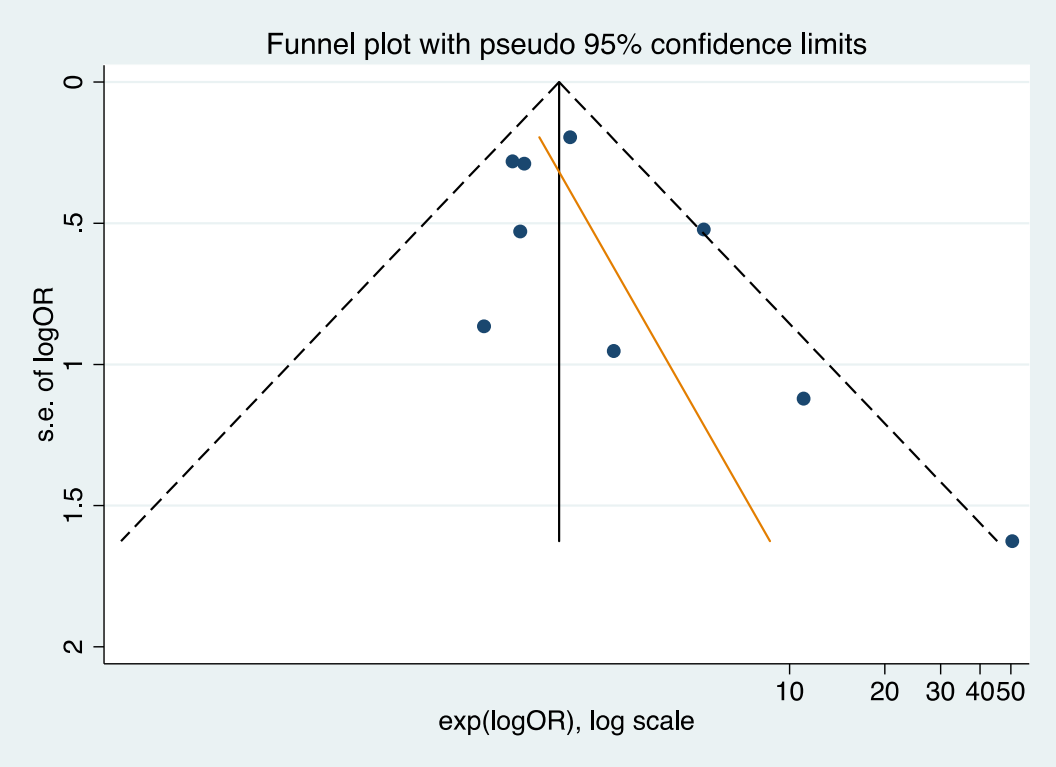


Figure 28; Diabetes

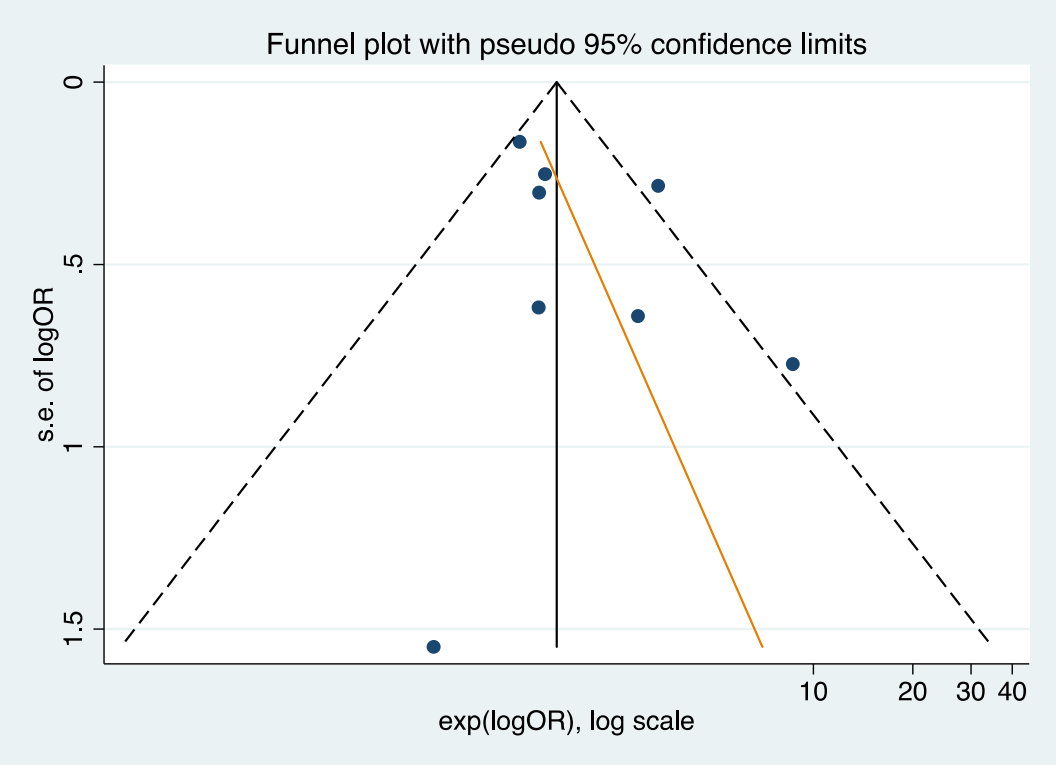


Figure 29; Hypertension

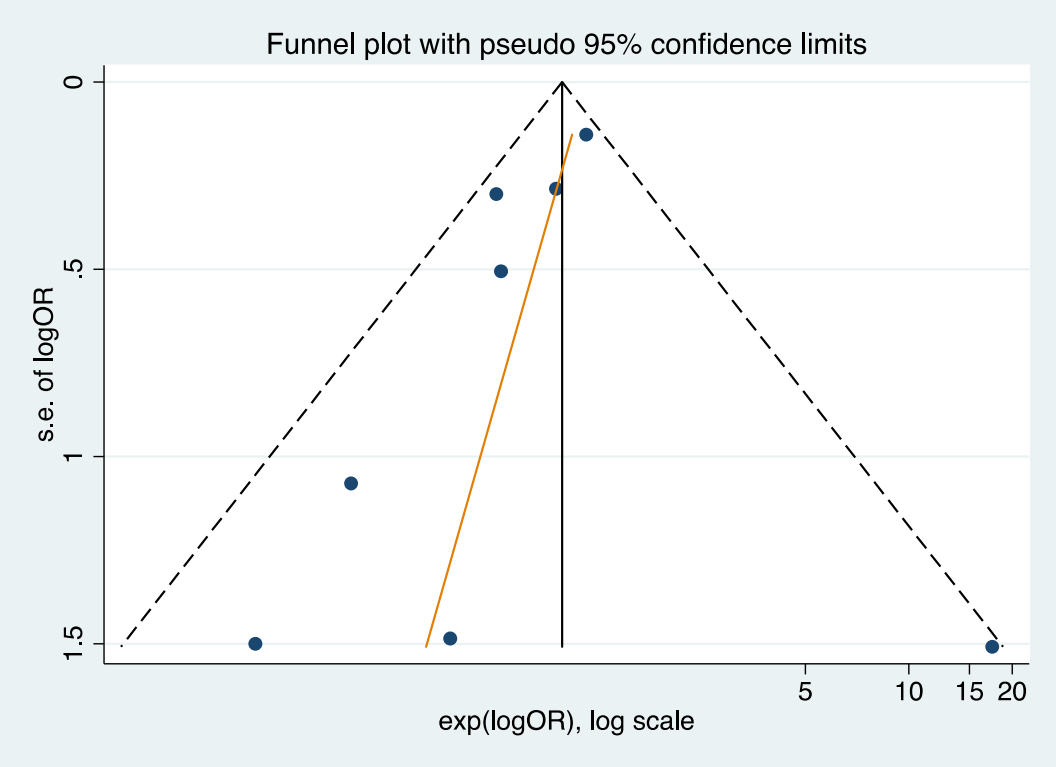


Figure 30; Smoking

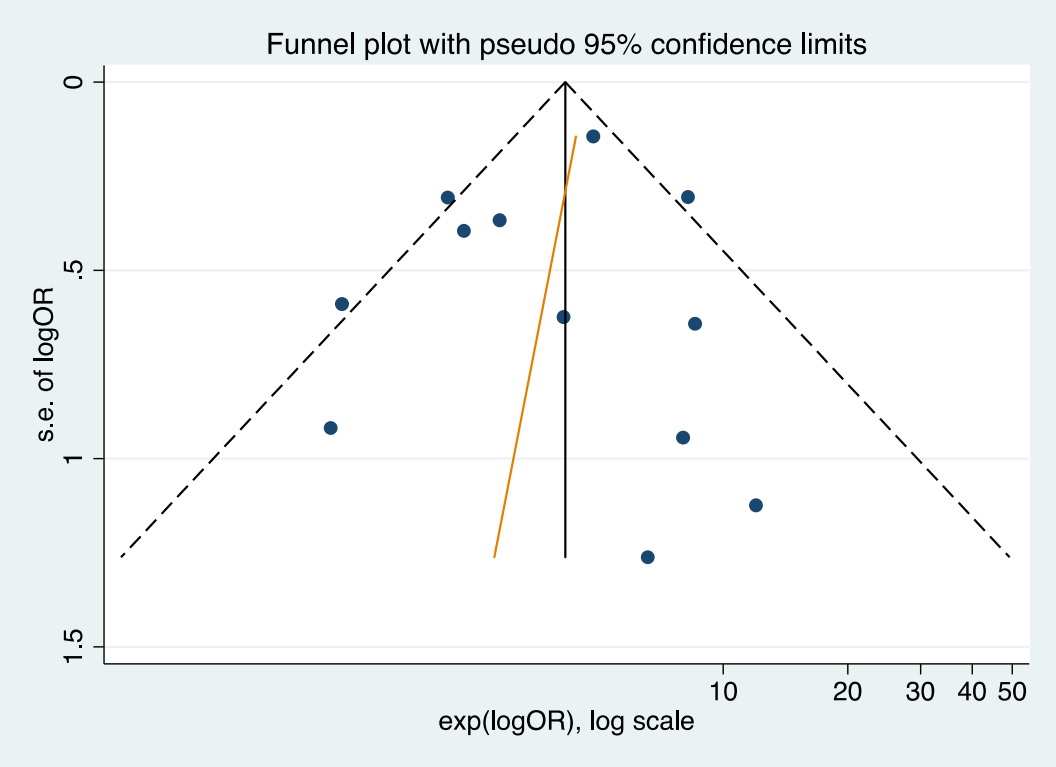


Figure 31; Progressive MS

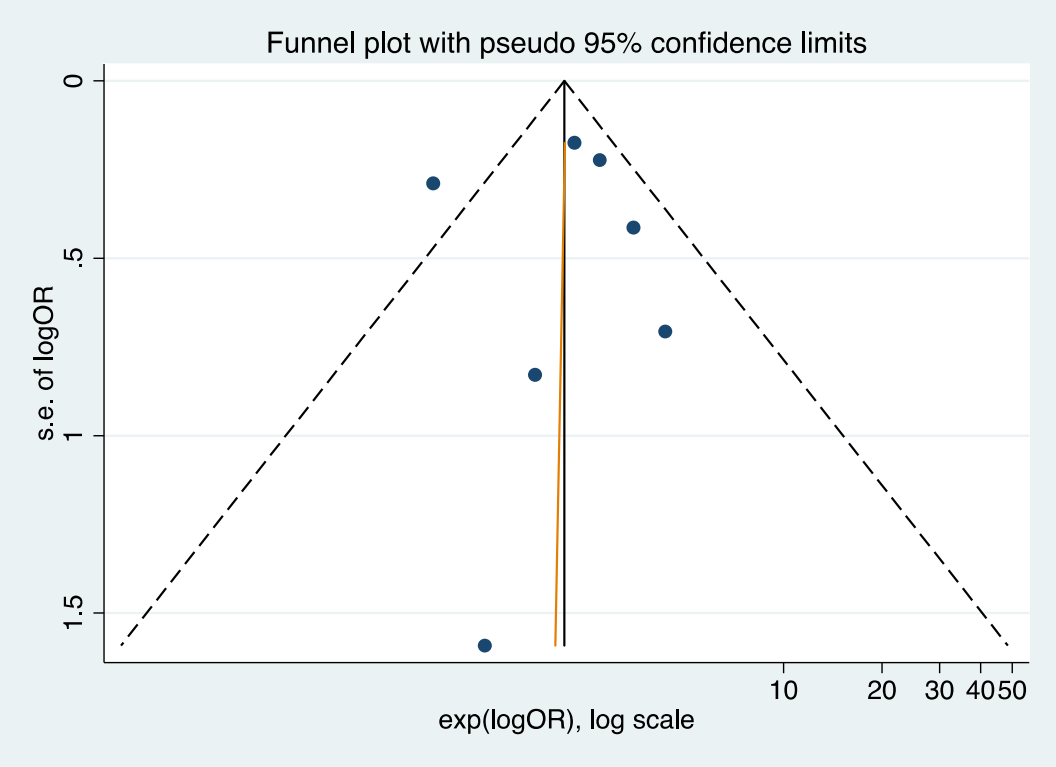


Figure 32; 3<EDSS<6

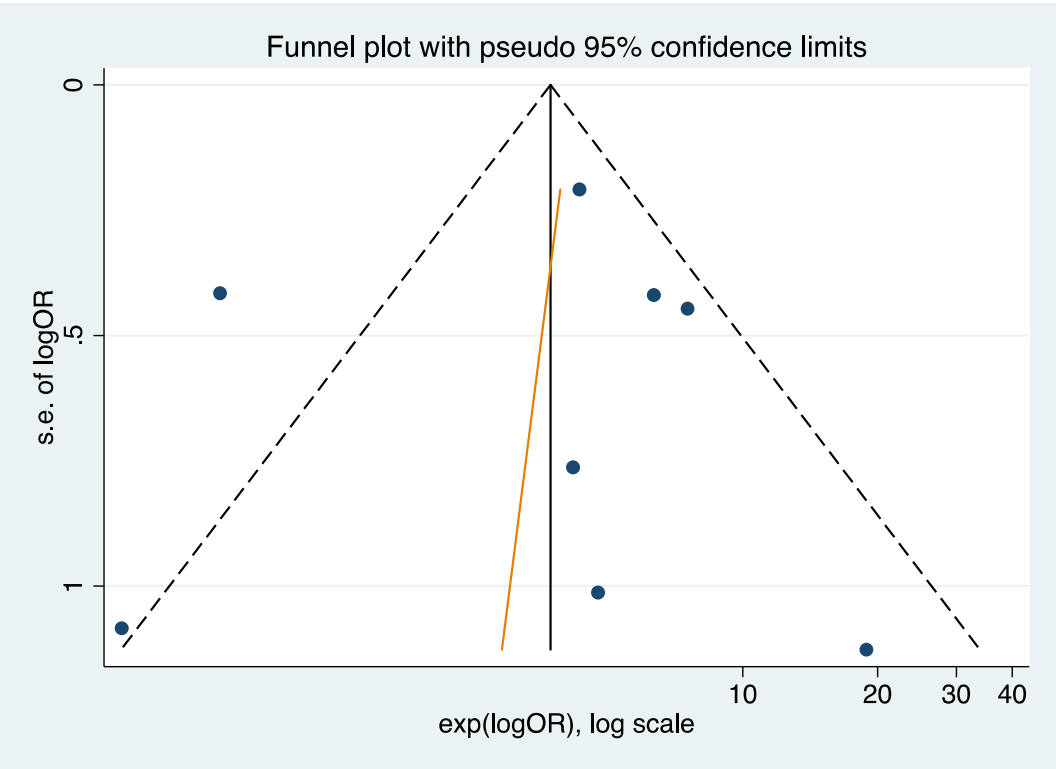


Figure 33; EDSS>6

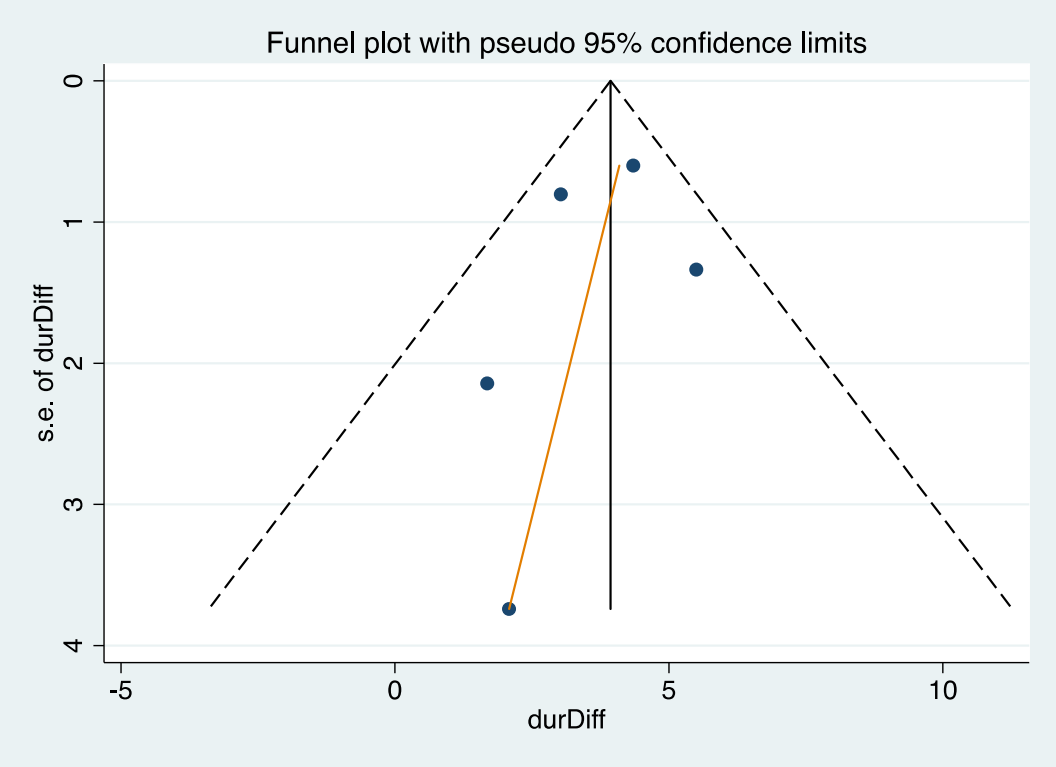


Figure 34; Disease duration

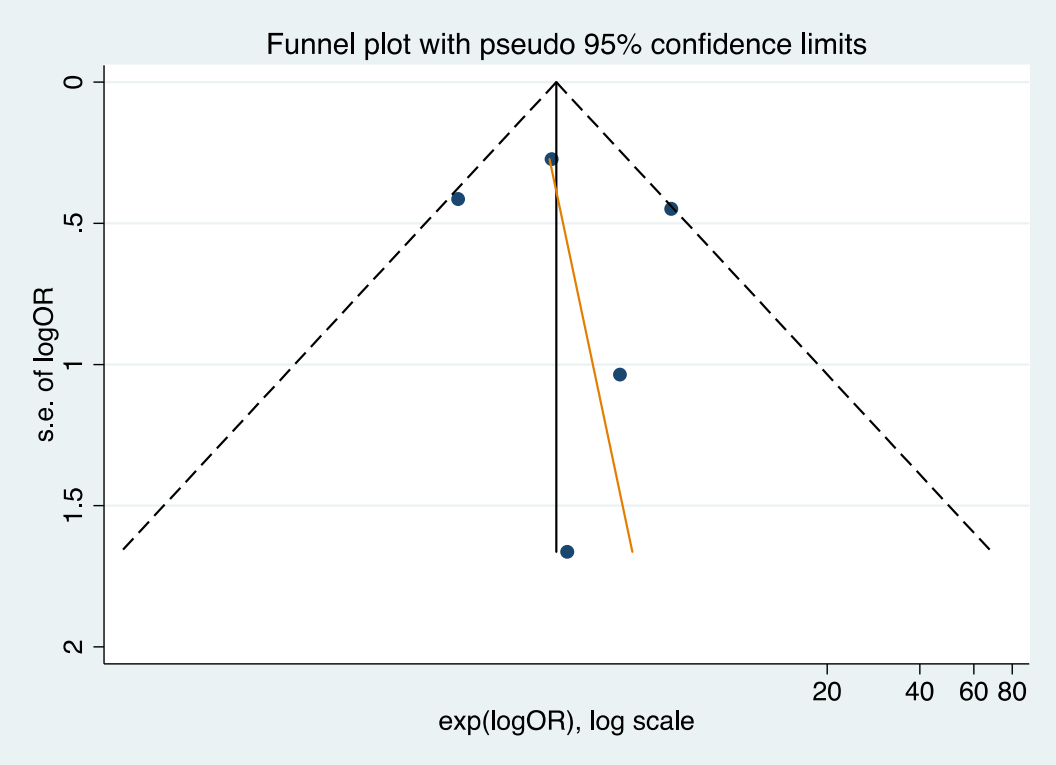


Figure 35; Corticosteroid use within 2 months

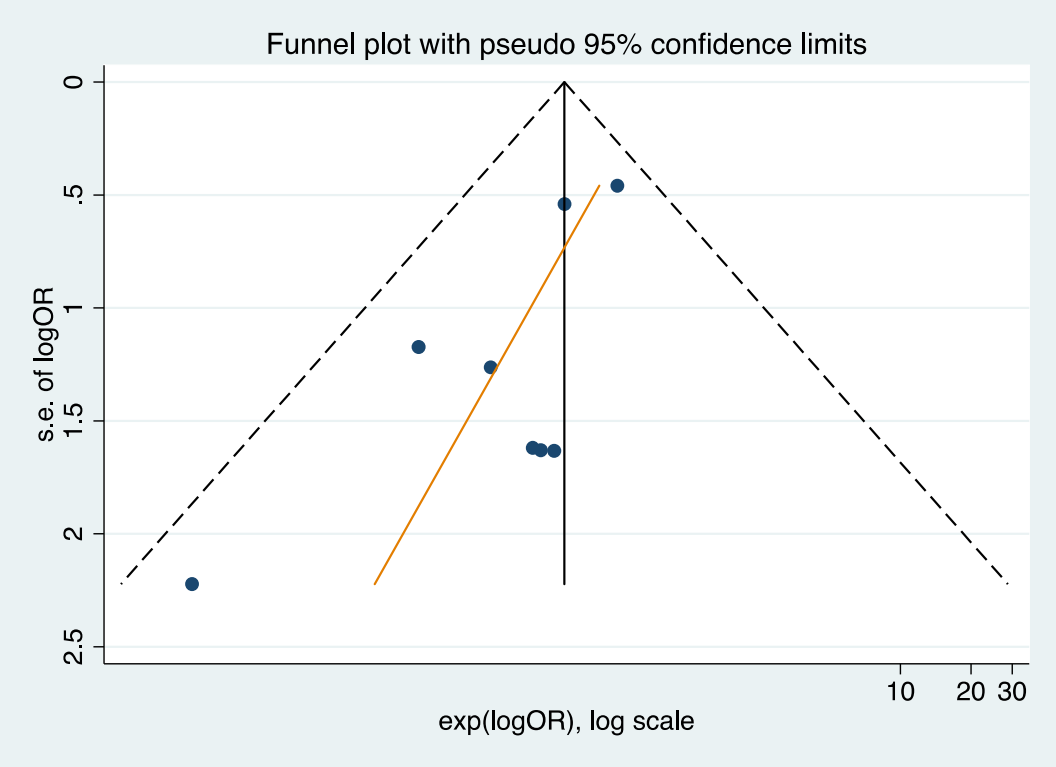


Figure 36; Interferons

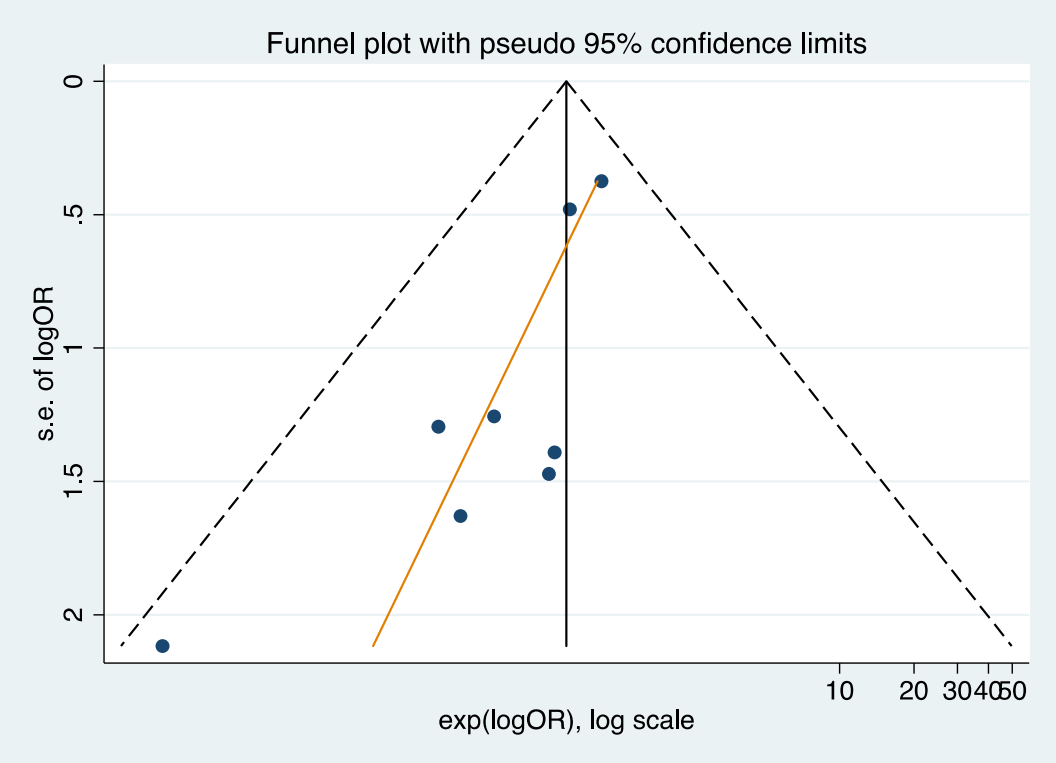


Figure 37; Glatiramer

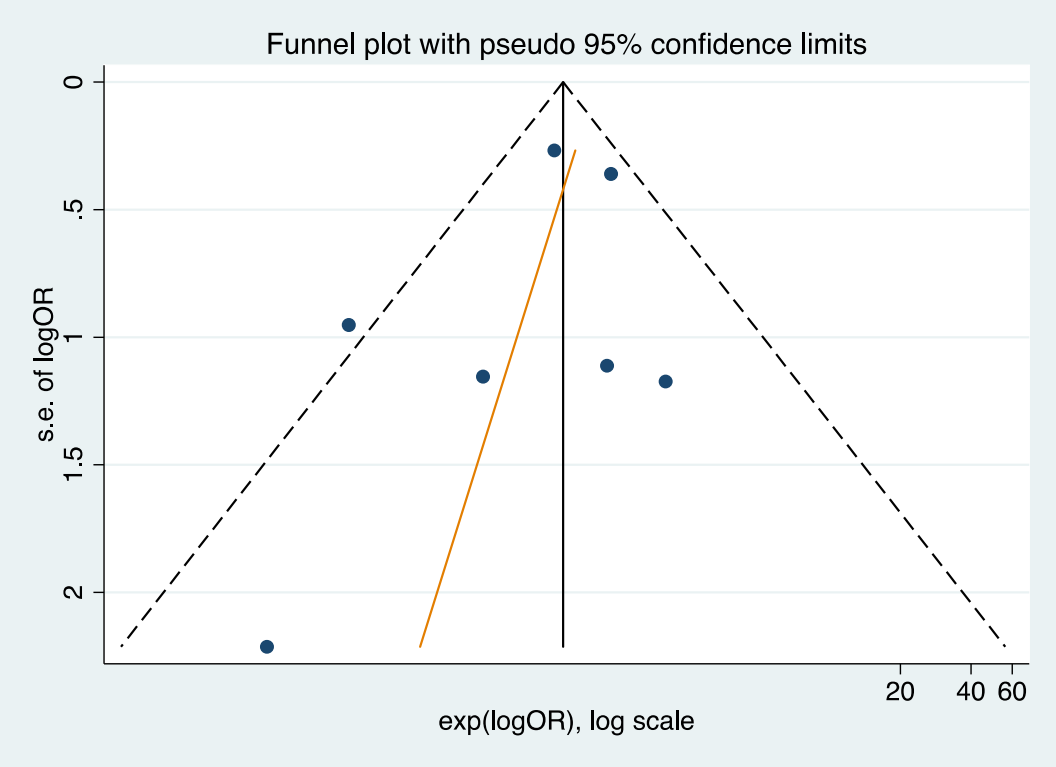


Figure 38; Fumarate

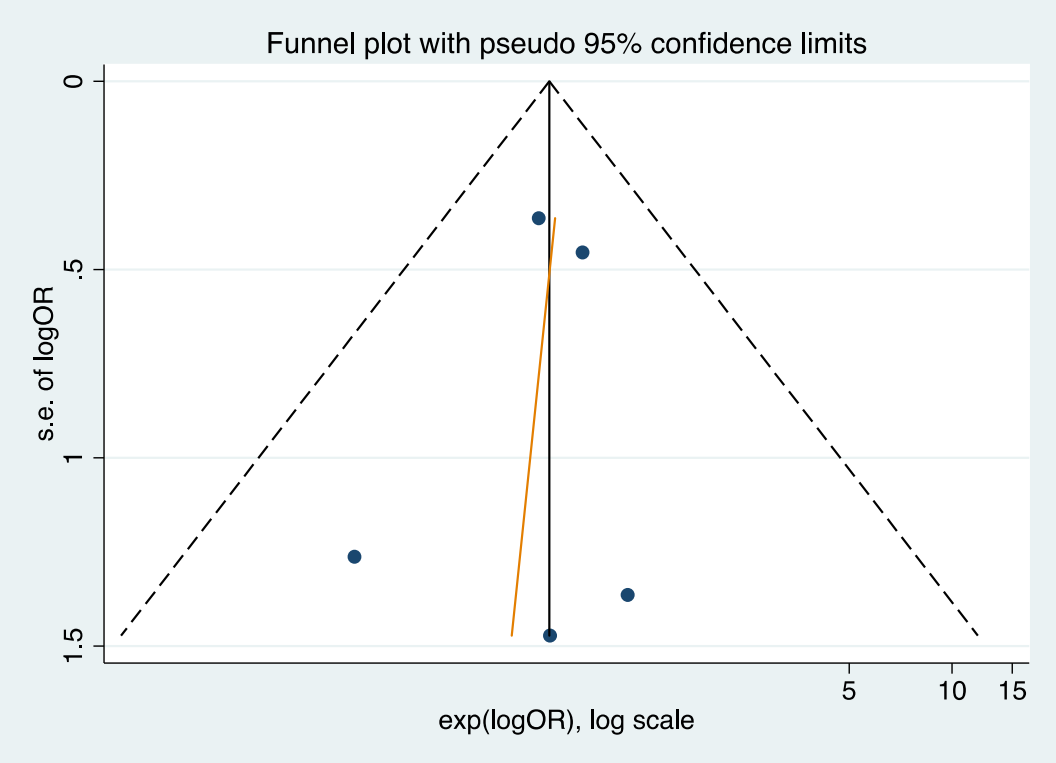


Figure 39; Teriflunomide

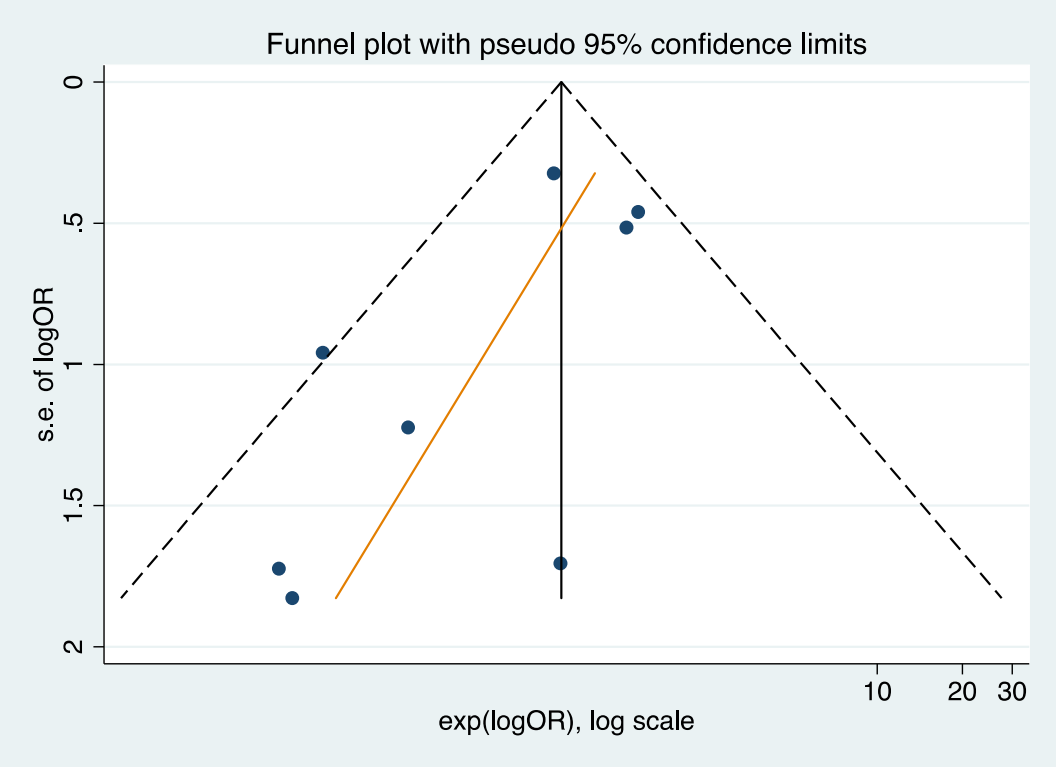


Figure 40; Fingolimod

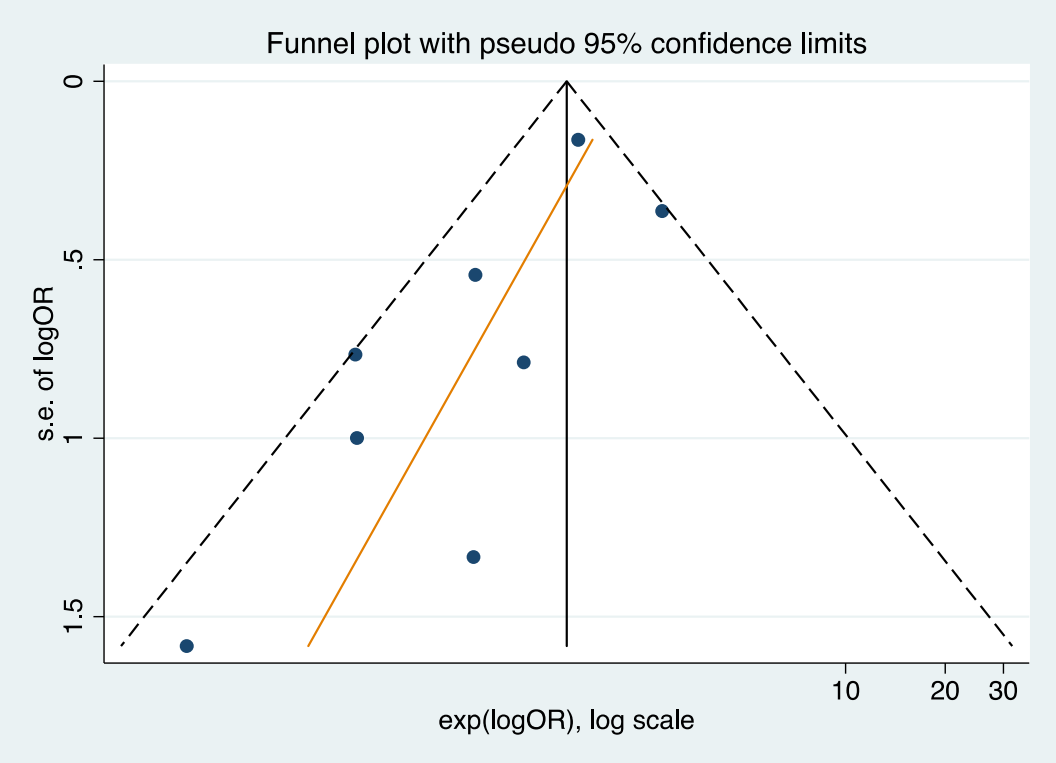


Figure 41; Anti-CD20 therapies

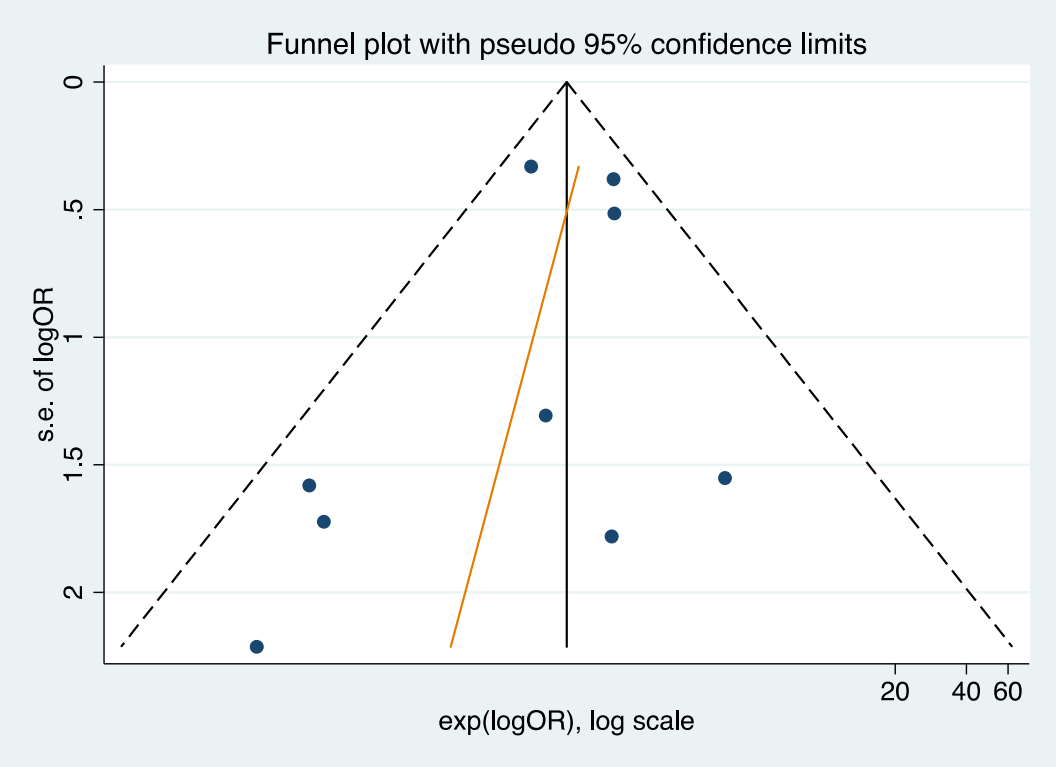


Figure 42; Natalizumab

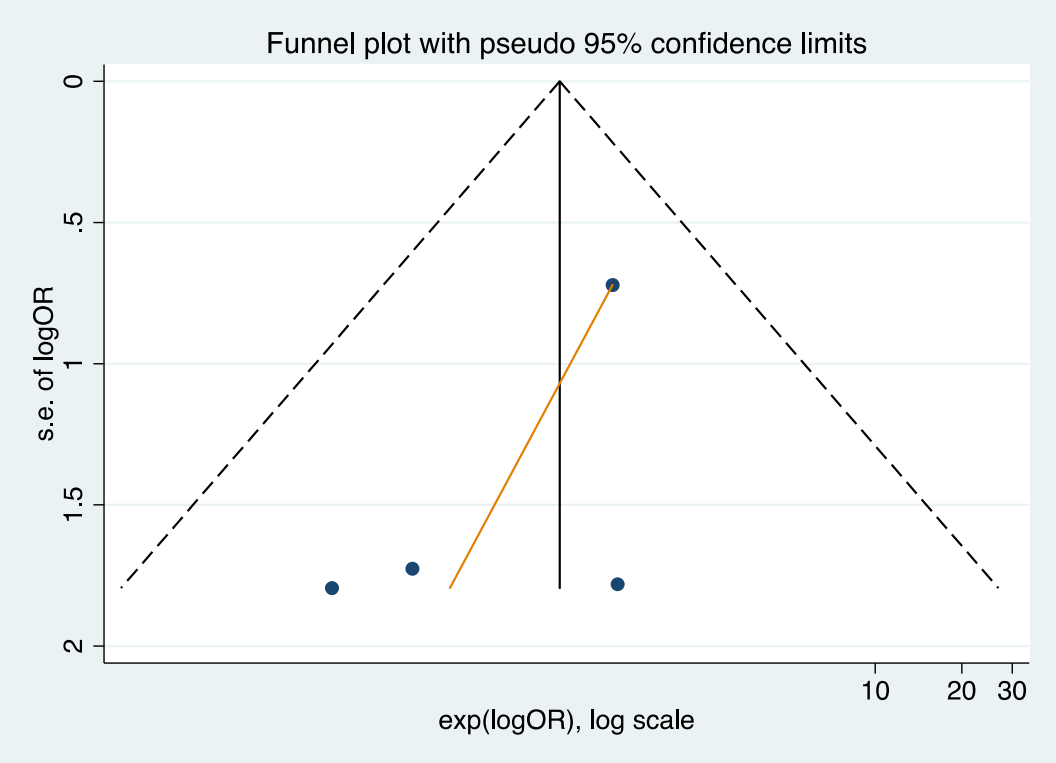


Figure 43; Alemtuzumab

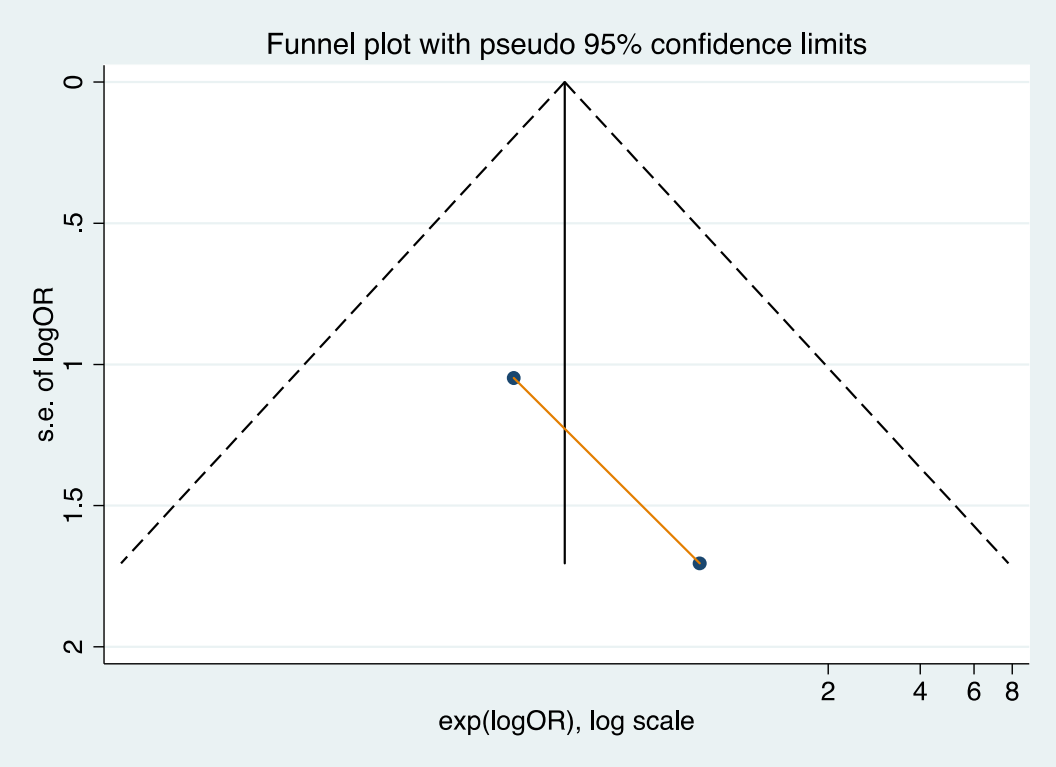


Figure 44; Cladribine

4. Sensitivity analysis

Table 1; results of sensitivity analysis

Risk Factors	Meta-analysis of studies with good quality (less bias)	Egger's test
	Effect size (95% CI)	P value
Overall demographics		
Age	1.58 (1.29, 1.94)	0.375
Male sex	1.44 (1.15, 1.80)	0.039
Comorbidities		
Obesity	1.69 (0.86, 3.31)	0.404
Diabetes	2.17 (1.55, 3.05)	0.151
Pulmonary comorbidities	1.70 (1.12, 2.57)	0.393
Cardiovascular comorbidities	1.94 (1.30, 2.90)	0.490
Hypertension	1.33 (1.00, 1.77)	0.284
Current/past Smoking	1.00 (0.79, 1.28)	0.333
MS characteristics		
Progressive MS	4.99 (3.12, 7.98)	0.595
3<EDSS<6	1.85 (0.87, 3.91)	0.969
EDSS>6	2.70 (0.83, 8.78)	0.827
Disease duration	3.88 (2.93, 4.82)	0.574

Corticosteroid in past 2 months	3.70 (1.56, 8.81)	0.769
DMTs		
Interferons	0.50 (0.25, 0.99)	0.029
Glatiramer	0.97 (0.55, 1.74)	0.012
Teriflunomide	0.69 (0.40, 1.21)	0.686
Fumarate	0.82 (0.54, 1.25)	0.377
Fingolimod	0.90 (0.54, 1.52)	0.083
Anti-CD20	1.72 (1.28, 2.30)	0.058
Natalizumab	0.82 (0.50, 1.33)	0.331
Alemtuzumab	1.22 (0.30, 5.01)	0.265
Cladribine	0.19 (0.02, 1.46)	NA

5. Meta-Regression Results

Table 2; results of meta-regression

Variable	B	LCI	UCI	P
Age	.2585905	-.4314912	.9486721	0.405
Obesity	-.2435873	-1.58821	1.101036	0.673
Hypertension	-.555425	-1.358905	.2480554	0.142
Progressive MS	-.1619571	-1.150127	.8262127	0.723
3<EDSS<6	-.5030489	-2.556233	1.550135	0.556
EDSS>6	-.8936632	-3.147973	1.360647	0.370